

Science Olympiad
Science Olympiad at Berkeley Invitational
February 28, 2026

Machines B Answer Key



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Section A (40 points)

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| 1. <u> E </u> | 2. <u> B </u> | 3. <u> B </u> | 4. <u> E </u> | 5. <u> C </u> |
| 6. <u> B </u> | 7. <u> B </u> | 8. <u> A </u> | 9. <u> D </u> | 10. <u> A </u> |
| 11. <u> A </u> | 12. <u> B </u> | 13. <u> B </u> | 14. <u> B </u> | 15. <u> D </u> |
| 16. <u> A </u> | 17. <u> D </u> | 18. <u> C </u> | 19. <u> C </u> | 20. <u> E </u> |
| 21. <u> C </u> | 22. <u> D </u> | 23. <u> E </u> | 24. <u> C </u> | 25. <u> E </u> |
| 26. <u> C </u> | 27. <u> D </u> | 28. <u> D </u> | | |

Section B (50 points)

1. [1 pt] Pulley
2. [2 pts] Ideal: 80 N (+1), Actual: 160 N (+1)
3. [1 pt] 393
4. [1 pt] Input work lost as thermal energy so can't contribute to useful output work. (+1 for mentioning loss of energy)
5. [2 pts] thread depth = 1.3 mm (+1), minor diameter = 9.4 mm (+1)
6. [2 pts] Effort arm is shorter than load arm (+1), so requires larger effort force than load (+1)
7. [3 pts] IMA = 2 (+1), AMA = 1.49 (+1), efficiency = 74% (+1)
8. [1 pt] Wedge/inclined plane
9. [2 pts] AMA = 5 (+0.5), IMA = 6 (+0.5), efficiency = 83.3% (+1)
10. [2 pts] $X * Y * Z$ (+2)
11. [2 pts] 20 lbs (+2)
12. [2 pts] Bevel gear system (+1), planetary gear system (+1)
13. [2 pts] $\frac{151}{324}MR^2$
14. [2 pts] actual mechanical advantage = $6000/1500 = 4$ (+0.5), IMA = $1/(\tan(9^\circ)) = 6.31$ (+0.5). Efficiency = AMA/IMA (+0.5) = $4/6.31 = 0.634 = 63.4\%$ (+0.5)

15. [12 pts] (a) [2 pts] $\frac{2R}{R-r}$ (+1), $IMA = 10$ (+1)
- (b) [2 pts] $F_e = F_l * IMA$ (+1), 98 N (+1)
- (c) [2 pts] $AMA = 10 * 0.6 = 6$ (+1), $Load = 150 * 6 = 900$ N (+1)
- (d) [3 pts] $V_{load} = V_e / IMA = 0.05 m/s$ (+1), $P_{out} = F_{load} * V_{load}$ (+1), 24.5 Watts (+1)
- (e) [2 pts] $\Delta PE = mgh$ (+1), 980 J (+1)
- (f) [1 pt] Less than g
16. [13 pts] (a) [2 pts] $IMA_{handle} = \frac{10}{7} = 1.43$ (+1), $IMA_{total} = 1.43 * 0.15 = 0.215$ (+1)
- (b) [3 pts] Barely touching = maximum distance between 2 while touching = $2 * radius = 30$ cm (+1).
Distance between = 4 m. $\sin(stone1) = \frac{30}{400} = 4.3^\circ$ (+1). Final angle = $20 - 4.3 = 15.7^\circ$ (+1)
- (c) [3 pts] At time of contact, stone 1 needs to be horizontal with the second puck. projection of 1 to 2 in the horizontal direction = $4 * \cos(20^\circ) - 0.3 = 3.46m$ (+1), vertical projection = $4 * \sin(20^\circ) = 1.37m$ (+1). $\tan(angle) = \frac{1.37}{3.46} = 21.6^\circ$ (+1)
- (d) [2 pts] Law of sines (+1), $\frac{4}{\sin(20^\circ)} = \frac{5}{\sin(x)} \implies x = 25.31^\circ$ (+1)
- (e) [3 pts] Use law of cosines to find the distance to the center of the target: 8.57 m (+1). Minimum distance needed to travel: $8.57 - (2/2) + (0.3/2) = 7.72$ m. (+0.5), deceleration of the puck is - $0.588 m/s^2$ (+0.5), $v_f = v_i^2 + 2a\Delta x$ (+0.5), solving: $v_f = \sqrt{2 * 0.588 * 7.72} = 3.01 m/s$ (+0.5).
 $F_{handle} = mv / \Delta t = 18 * 3.01 / 1 = 54.18 N$